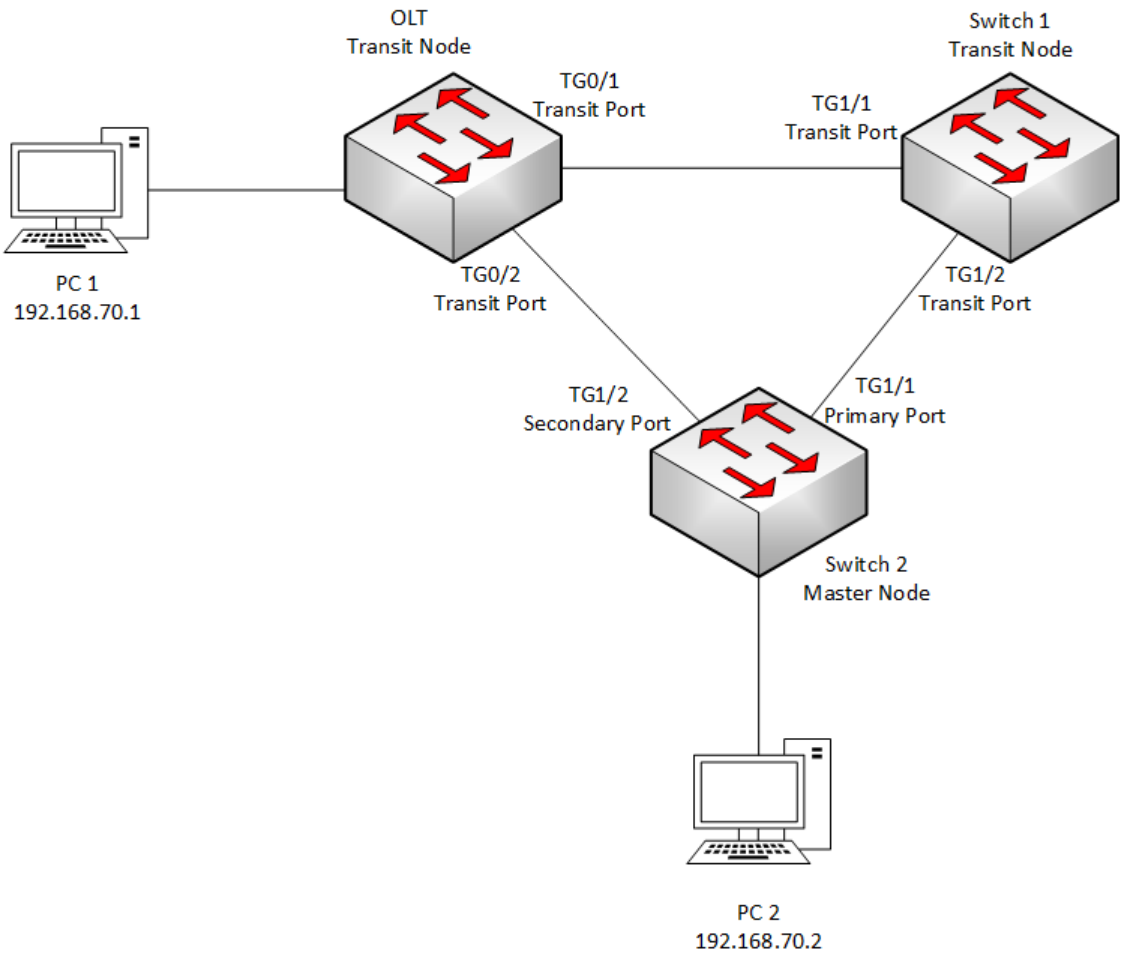
Test Report	**Test report on EAPS through BDCOM devices**
Objective	EAPS stands for Ether Automatic Protection Switching which is applied to a ring network topology to ensure the redundant and in this configuration traffic flow can be defined by the user
Testing environment	Lab environment at BDCOM Asia Pacific Office, Dhaka, Bangladesh
Tested By	Salakin, After Sales Engineer
Date	20th November 2019
Testing devices	1. OLT Model: P3608-2TE Firmware version: 10.1.0E Build 66506 2. Switch1 Model: S3756F Firmware version: 2.2.0C Build 68373 3. Switch2 Model: S3756 Firmware version: 2.2.0C Build 63122

Topology:



Configuration:

OLT (Transit Node):

```
OLT_config#vlan 10
OLT_config_v10#exit
OLT_config#no spanning-tree
OLT_config#ether-ring 1
OLT_config_ring1#control-vlan 10
OLT_config_ring1#transit-node
OLT_config_ring1#exit
OLT_config#interface TGigaEthernet0/1        // Connected with SW1 in TG1/1
OLT_config_tg0/1#switchport mode trunk
OLT_config_tg0/1#ether-ring 1 transit-port
OLT_config_tg0/1#exit
OLT_config#interface TGigaEthernet0/2        // Connected with SW2 in TG1/2
OLT_config_tg0/2#switchport mode trunk
OLT_config_tg0/2#ether-ring 1 transit-port
OLT_config_tg0/2#exit
```

Switch 1 (Transit Node):

```
SW1_config#vlan 10
SW1_config_v10#exit
SW1_config#no spanning-tree
SW1_config#ether-ring 1
SW1_config_ring1#control-vlan 10
SW1_config_ring1#transit-node
SW1_config_ring1#exit
SW1_config#interface TGigaEthernet1/1        // Connected with OLT in TG0/1
SW1_config_tg1/1#switchport mode trunk
SW1_config_tg1/1#ether-ring 1 transit-port
SW1_config_tg1/1#exit
SW1_config#interface TGigaEthernet1/2        // Connected with SW2 in TG1/2
SW1_config_tg1/2#switchport mode trunk
SW1_config_tg1/2#ether-ring 1 transit-port
SW1_config_tg1/2#exit
```

Switch 2 (Master Node):

```
SW2_config#vlan 10
SW2_config_v10#exit
SW2_config#no spanning-tree
SW2_config#ether-ring 1
SW2_config_ring1#control-vlan 10
SW2_config_ring1#master-node
SW2_config_ring1#exit
SW2_config#interface TGigaEthernet1/1        // Connected with SW1 in TG1/2
SW2_config_tg1/1#switchport mode trunk
SW2_config_tg1/1#ether-ring 1 primary-port
SW2_config_tg1/1#exit
SW2_config_tg1/2#interface TGigaEthernet1/2   // Connected with OLT in TG0/2
SW2_config_tg1/2#switchport mode trunk
SW2_config_tg1/2#ether-ring 1 secondary-port
SW2_config_tg1/2#exit
```

Results: (When all links are up)

SW2 (Master Node):

SW2#**show ether-ring 1**

```
Ethernet Automatic Protection Switching

ether-ring 1
  Configured   Role          Master-node
               Control-Vlan   10
               Node ID        8479.738F.AE9E
               Hello Time 1 sec Fail Time 3 sec Pre-forward
 Time 3 sec

  Running      State         Idle
               Mode          Independent
               Complete      True
               Health Check   True
               Hello While 1 sec Fail While 3 sec

Interface       Role        State       Pre-fwd Status
---------------- ---------- ------------ -------- ------------------------
tg1/1           Primary     FWD          0        Enabled, Link-Up
tg1/2           Secondary   BLK          0        Enabled, Link-Up
```

SW1 (Transit Node):

SW1#**show ether-ring 1**

```
Ethernet Automatic Protection Switching

ether-ring 1
  Configured   Role          Transit-node
               Control-Vlan   10
               Node ID        8479.735F.6951
               Hello Time 1 sec Fail Time 3 sec Pre-forward
Time 3 sec

  Running      State         Links-Up
               Mode          Independent
               Master ID      8479.738F.AE9E
               Complete      True

Interface       Role       State       Pre-fwd Status
---------------- ---------- ------------ -------- ------------------------
tg1/1           Transit    FWD          0        Enabled, Link-Up
tg1/2           Transit    FWD          0        Enabled, Link-Up
```

OLT (Transit Node):

OLT# **show ether-ring 1**

Ethernet Automatic Protection Switching

ether-ring 1
Configured Role Transit-node
Control-Vlan 10
Node ID 8479.73DF.17AD
Hello Time 1 sec Fail Time 3 sec Pre-forward Time 3 sec

Running State Links-Up
Mode Independent
Master ID 8479.738F.AE9E
Complete True

Interface	Role	State	Pre-fwd	Status
tg0/1	Transit	FWD	0	Enabled, Link-Up
tg0/2	Transit	FWD	0	Enabled, Link-Up

Results:(When link between SW1 and OLT is down)

SW2 (Master Node):

1 1 06:17:42 %EAPS-4: ether-ring 1 tg1/2 received Link-Down from node system 8479.73DF.17B7
1 1 06:17:42 %EAPS-4: ether-ring 1 Secondary Port tg1/2 Forward
1 1 06:17:42 %EAPS-4: ether-ring 1 Ring-Fault
1 1 06:17:42 %EAPS-4: ether-ring 1 tg1/1 received Link-Down from node system 8479.735F.6983

SW2#**show ether-ring 1**

Ethernet Automatic Protection Switching

ether-ring 1
Configured Role Master-node
Control-Vlan 10
Node ID 8479.738F.AE9E
Hello Time 1 sec Fail Time 3 sec Pre-forward
Time 3 sec

Running State Idle
Mode Independent
Complete False
Health Check True
Hello While 1 sec Fail While 0 sec

Interface	Role	State	Pre-fwd	Status
tg1/1	Primary	FWD	0	Enabled, Link-Up
tg1/2	Secondary	FWD	0	Enabled, Link-Up

SW1 (Transit Node):

Jan 1 06:28:27 %LINE-5-UPDOWN: Line on Interface TGigaEthernet1/1, changed state to down
Jan 1 06:28:27 %LINEPROTO-5-UPDOWN: Line protocol on Interface TGigaEthernet1/1, changed state to down

SW1#**show ether-ring 1**

Ethernet Automatic Protection Switching

```
ether-ring 1
  Configured   Role          Transit-node
               Control-Vlan   10
               Node ID        8479.735F.6951
               Hello Time 1 sec Fail Time 3 sec Pre-forward Time 3 sec

  Running      State         Link-Down
               Mode          Independent
               Master ID      8479.738F.AE9E
               Complete       False

Interface        Role       State        Pre-fwd Status
---------------- ---------- ------------ -------- -----------------------
tg1/1            Transit    BLK          0        Enabled, Link-Down
tg1/2            Transit    FWD          0        Enabled, Link-Up
```

From OLT (Transit Node):

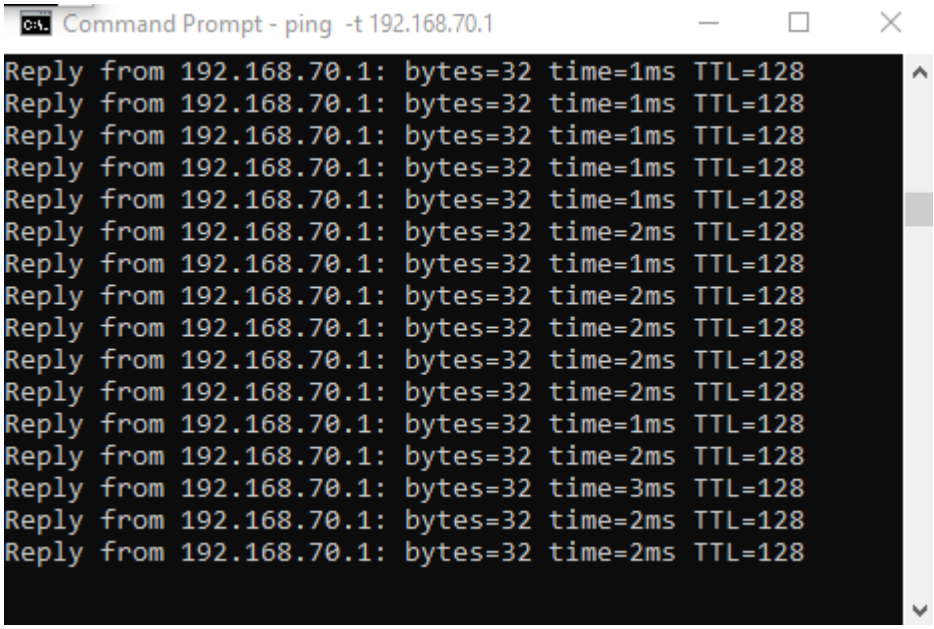
Switch#Jan 1 02:34:15 %LINE-5-UPDOWN: Line on Interface TGigaEthernet0/1, changed state to down
Jan 1 02:34:15 %LINEPROTO-5-UPDOWN: Line protocol on Interface TGigaEthernet0/1, changed state to down

Switch# **show ether-ring 1**

Ethernet Automatic Protection Switching

```
ether-ring 1
  Configured   Role          Transit-node
               Control-Vlan   10
               Node ID        8479.73DF.17AD
               Hello Time 1 sec Fail Time 3 sec Pre-forward Time 3 sec

  Running      State         Link-Down
               Mode          Independent
               Master ID      8479.738F.AE9E
               Complete       False

Interface        Role       State        Pre-fwd Status
---------------- ---------- ------------ -------- -----------------------
tg0/1            Transit    BLK          0        Enabled, Link-Down
tg0/2            Transit    FWD          0        Enabled, Link-Up
```

Ping snap during link swap,